

Part A: Complete each question, showing work in the space provided. Do not attach additional pages. Each question on this page is worth 5 pts.

1. Use the inner product $\langle p, q \rangle = a_0 b_0 + a_1 b_1 + a_2 b_2$ to find $\langle p, q \rangle$, $\|p\|$, $\|q\|$, and $d(p, q)$.

$$p(x) = 1 - x + 4x^2, q(x) = x - x^2$$

$$1.) p(x) = 1 - x + 4x^2 \rightarrow p = (1, -1, 4)$$

$$q(x) = x - x^2 \rightarrow q = (0, 1, -1)$$

$$2.) \langle p, q \rangle = 1(0) + (-1)(1) + 4(-1) = 0 - 1 - 4 = -5$$

$$3.) \|p\| = \sqrt{\langle p, p \rangle} = \sqrt{(1)^2 + (-1)^2 + (4)^2} = \sqrt{18} = 3\sqrt{2}$$

$$\|q\| = \sqrt{\langle q, q \rangle} = \sqrt{0^2 + (1)^2 + (-1)^2} = \sqrt{2}$$

$$4.) d(p, q) = \|p - q\|$$

$$p - q = (1, -1, 4) - (0, 1, -1) = (1, -2, 5)$$

$$d(p, q) = \sqrt{(1)^2 + (-2)^2 + (5)^2}$$

$$= \sqrt{1 + 4 + 25}$$

$$= \sqrt{30}$$

2. Find the projection of the vector $v = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}$ onto the subspace $S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$

$v_1 \quad v_2$

$$1.) \text{Proj}_S(v) = (v \cdot u_1)u_1 + (v \cdot u_2)u_2$$

$v_1 \cdot v_2 = 1 \neq 0$ Not orthogonal, so we must make an orthonormal basis using Gram-Schmidt

$$2.) u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$3.) w_2 = v_2 - (v_2 \cdot u_1)u_1$$

$$v_2 \cdot u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

$$w_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

$$\|w_2\| = \sqrt{1/4 + 1/4 + 1} = \sqrt{3/2} = \sqrt{6}/2$$

$$u_2 = \frac{w_2}{\|w_2\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

$$4.) v \cdot u_1 = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \frac{4+2}{\sqrt{2}} = 3\sqrt{2}$$

$$(v \cdot u_1)u_1 = 3\sqrt{2} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix}$$

$$v \cdot u_2 = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix} \cdot \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \frac{4-2+6}{\sqrt{6}} = \frac{8}{\sqrt{6}}$$

$$(v \cdot u_2)u_2 = \frac{8}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \frac{8}{6} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4/3 \\ -4/3 \\ 8/3 \end{bmatrix}$$

$$\text{Proj}_S(v) = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix} + \begin{bmatrix} 4/3 \\ -4/3 \\ 8/3 \end{bmatrix} = \begin{bmatrix} 13/3 \\ 5/3 \\ 8/3 \end{bmatrix}$$

3. Let θ be the angle between vectors $\mathbf{u} = (4, -3)$, $\mathbf{v} = (0, 5)$. Find $\cos \theta$ and θ .

$$\cos \theta = \frac{(\mathbf{u} \cdot \mathbf{v})}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

$$1.) \mathbf{u} \cdot \mathbf{v} = 4(0) + (-3)(5) = -15$$

$$3.) \cos \theta = \frac{-15}{5 \cdot 5} = -\frac{3}{5}$$

$$2.) \|\mathbf{u}\| = \sqrt{4^2 + (-3)^2} = \sqrt{16+9} = 5$$

$$4.) \theta = \cos^{-1}\left(-\frac{3}{5}\right)$$

$$\|\mathbf{v}\| = \sqrt{0^2 + 5^2} = 5$$

$$\theta \approx 2.214 \text{ radians}$$

4. Find $\mathbf{u} \cdot \mathbf{v}$, $\mathbf{v} \cdot \mathbf{v}$, $\|\mathbf{u}\|^2$, $(\mathbf{u} \cdot \mathbf{v})\mathbf{v}$, and $\mathbf{u} \cdot (5\mathbf{v})$

$$\mathbf{u} = (5, 2, 0, 4), \mathbf{v} = (0, -3, 4, 5)$$

$$1.) \mathbf{u} \cdot \mathbf{v} = 5(0) + 2(-3) + 0(4) + 4(5) = 0 - 6 + 0 + 20 = 14$$

$$2.) \mathbf{v} \cdot \mathbf{v} = (0)^2 + (-3)^2 + 4^2 + 5^2 = 0 + 9 + 16 + 25 = 50$$

$$3.) \|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u} = 5^2 + 2^2 + 0^2 + 4^2 = 25 + 4 + 0 + 16 = 45$$

$$4.) (\mathbf{u} \cdot \mathbf{v})\mathbf{v} = 14(0, -3, 4, 5) = (0, -42, 56, 70)$$

$$5.) \mathbf{u} \cdot (5\mathbf{v}): \quad 5\mathbf{v} = (0, -15, 20, 25)$$

$$\mathbf{u} \cdot (5\mathbf{v}) = 5(0) + 2(-15) + 0(20) + 4(25)$$

$$= 0 - 30 + 0 + 100 = 70$$

5. Consider vectors $u = (-4, \frac{7}{2}, -9, \frac{9}{2})$ and $v = (2, -\frac{7}{4}, \frac{9}{2}, -\frac{9}{4})$. Find the dot product of the two vectors, a scalar value c such that $u = cv$, and then determine whether u and v are orthogonal, parallel or neither. Explain.

$$1.) u \cdot v = (-4)(2) + (\frac{7}{2})(-\frac{7}{4}) + (-9)(\frac{9}{2}) + (\frac{9}{2})(-\frac{9}{4})$$

$$= -8 - \frac{49}{8} - \frac{81}{2} - \frac{81}{8} = \frac{-64 + 49 + 324 + 81}{8} = -\frac{518}{8} = -\frac{259}{4}$$

$$2.) u = cv: -4 = 2c \rightarrow c = -2$$

$$\text{Check other components; } \frac{7}{2} = (-2)(-\frac{7}{4}) = \frac{7}{2} \checkmark$$

$$-9 = (-2)(\frac{9}{2}) = -9 \checkmark$$

$$\frac{9}{2} = (-2)(-\frac{9}{4}) = \frac{9}{2} \checkmark$$

$$3.) \text{ Since } u = -2v, \text{ the vectors are}$$

Scalar multiples, therefore parallel.

They are not orthogonal because the dot

Product is not zero.

6. Define the linear transformation T by $T(x) = Ax$ for $A = \begin{bmatrix} 0 & -3 & 8 \\ 6 & 0 & 13 \end{bmatrix}$. Find $\ker(T)$ (using t as your parameter for an infinite number of solutions), $\text{nullity}(T)$, $\text{range}(T)$, and $\text{rank}(T)$

$$1.) \ker(T): \begin{bmatrix} 0 & -3 & 8 \\ 6 & 0 & 13 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 13/6 \\ 0 & 1 & -8/3 \end{bmatrix}, x_1 + \frac{13}{6}x_3 = 0, x_2 - \frac{8}{3}x_3 = 0, \text{ let } x_3 = t$$

$$\ker(T) = t \begin{bmatrix} -13/6 \\ 8/3 \\ 1 \end{bmatrix}$$

$$x_1 = -\frac{13}{6}t, x_2 = \frac{8}{3}t, x_3 = t$$

$$2.) \text{ Nullity}(T): \text{ There is one free variable } (x_3), \text{ so nullity}(T) = 1.$$

$$3.) \text{ Range}(T): \text{ The range is the column space of } A. \text{ Since the RREF has 2 pivots, rank}(T) = 2, \text{ so the range is all of } \mathbb{R}^2.$$

$$\text{range}(T) = \mathbb{R}^2$$

$$4.) \text{ rank}(T): \dim = \text{rank} + \text{nullity}$$

$$3 = \text{rank} + 1$$

$$\text{rank}(T) = 2$$

Part B: Application Questions (10 pts each). Choose 4 to solve. You may choose to solve an additional 1 question for up to 5 points of extra credit on this exam. Cross off the questions you don't want considered in your grade. Circle the question you want considered for extra credit.

For questions in Part B, I do NOT need to see row operations. Show the original matrix, then the row reduced matrix. Feel free to use your TI-84 or online calculator. You may also use the calculator to calculate the value of determinants in part B.



Find the least squares approximation $g(x) = a_0 + a_1x + a_2x^2$ of the function $f(x) = 4\sqrt{x}$, $1 \leq x \leq 4$.

8. Find the area of the parallelogram that has $\mathbf{u} = (3, -2, 0)$ and $\mathbf{v} = (-2, 4, 0)$ as adjacent sides.

$$\text{Area} = \|\mathbf{u} \times \mathbf{v}\|$$

$$1.) \mathbf{u} \times \mathbf{v} = \begin{bmatrix} i & j & k \\ 3 & -2 & 0 \\ -2 & 4 & 0 \end{bmatrix} = k(3 \cdot 4 - (-2)(-2)) = k(12 - 4) = 0i + 0j + 8k = (0, 0, 8)$$

$$2.) \|\mathbf{u} \times \mathbf{v}\| = \sqrt{0^2 + 0^2 + 8^2} = 8$$

$$\text{Area} = 8$$

9. Determine the matrix that produces the given pair of rotations.

60° about the z-axis followed by 45° about the y-axis. Then Find the image of the vector (1, 1, 1) under these rotations.

$$R = R_y(45^\circ) R_z(60^\circ)$$

$$R_y(45^\circ) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$1.) R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$2.) R = R_y R_z = \begin{bmatrix} \frac{\sqrt{2}}{4} & -\frac{\sqrt{6}}{4} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ -\frac{\sqrt{2}}{4} & \frac{\sqrt{6}}{4} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$R_z(60^\circ) = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3.) Image of (1, 1, 1):

$$\text{1st Component: } \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{2} = \frac{3\sqrt{2} - \sqrt{6}}{4}$$

$$\text{2nd Comp: } \frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{\sqrt{3} + 1}{2}$$

$$\text{3rd Comp: } -\frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{2} = \frac{\sqrt{2} + \sqrt{6}}{4}$$

$$\begin{bmatrix} \frac{3\sqrt{2} - \sqrt{6}}{4} \\ \frac{\sqrt{3} + 1}{2} \\ \frac{\sqrt{2} + \sqrt{6}}{4} \end{bmatrix}$$

10. Consider the linear transformation and basis $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2, T(x, y) = (3x - 4y, 2x)$, $B' = \{(-2, 1), (-1, 1)\}$. Find the standard matrix A for the linear transformation, the transition matrix p from B' to B, the inverse of p, and the matrix A' for T relative to the basis B' .

$$1.) \text{ Standard Matrix A for T: } A = \begin{bmatrix} 3 & -4 \\ 2 & 0 \end{bmatrix}, T(1, 0) = (3, 2), T(0, 1) = (-4, 0)$$

2. Transition Matrix P from B' to B: The transition matrix P from B' to the standard basis has the B' vectors as columns.

$$P = \begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix}$$

$$3.) P^{-1}: \det(P) = (-2)(1) - (-1)(1) = -2 + 1 = -1$$

$$P^{-1} = \frac{1}{-1} \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 1 & 2 \end{bmatrix}$$

$$4.) A' = P^{-1}AP: AP = \begin{bmatrix} 3 & -4 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -10 & -7 \\ -4 & -2 \end{bmatrix}$$

$$A' = P^{-1}(AP) = \begin{bmatrix} -1 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -10 & -7 \\ -4 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 9 \\ -18 & -11 \end{bmatrix}$$

11. The table shows the net profits (in millions of dollars) for Polo Ralph Lauren during the years 1996 to 2007.

Find the least squares regression line and the least squares cubic regression polynomial for the data.

Let t represent the year, with $t = 0$ corresponding to 2000. (Round each coefficient to four significant digits.)

Year	2000	2001	2002	2003	2004	2005	2006	2007
Net Profit	166.3	168.6	183.7	184.4	257.2	308.0	385.0	415.0

$$A^T A x = A^T b, \quad y = C_0 + C_1 t$$

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \end{bmatrix}, \quad x = \begin{bmatrix} C_0 \\ C_1 \end{bmatrix}, \quad b = \begin{bmatrix} 166.3 \\ 168.6 \\ 183.7 \\ 184.4 \\ 257.2 \\ 308.0 \\ 385.0 \\ 415.0 \end{bmatrix}$$

$$1.) A^T A = \begin{bmatrix} 8 & 28 \\ 28 & 140 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 2068.2 \\ 8873 \end{bmatrix}$$

$$2.) A^T A x = A^T b: \begin{bmatrix} 8 & 28 & | & 2068.2 \\ 28 & 140 & | & 8873 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & | & 122.3333333 \\ 0 & 1 & | & 38.91176476 \end{bmatrix}$$

$$x = \begin{bmatrix} C_0 \\ C_1 \end{bmatrix} = \begin{bmatrix} 122.33 \\ 38.91 \end{bmatrix}, \quad y \approx 122.33 + 38.91t$$

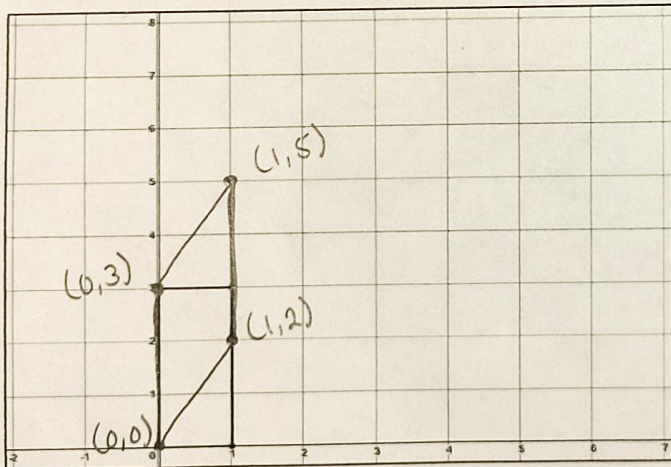
$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \end{bmatrix}, \quad x = \begin{bmatrix} C_0 \\ C_1 \\ C_2 \end{bmatrix}, \quad b = \text{Same as above}$$

$$y \approx C_0 + C_1 t + C_2 t^2$$

$$1.) A^T A = \begin{bmatrix} 8 & 28 & 140 \\ 28 & 140 & 784 \\ 140 & 784 & 4676 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 2068.2 \\ 8873 \\ 48573.2 \end{bmatrix}, \quad y \approx 161.5 - 0.24t + 5.59t^2$$

$$2.) \begin{bmatrix} 8 & 28 & 140 & | & 2068.2 \\ 28 & 140 & 784 & | & 8873 \\ 140 & 784 & 4676 & | & 48573.2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & | & 161.4833 \\ 0 & 1 & 0 & | & -0.2381 \\ 0 & 0 & 1 & | & 5.5929 \end{bmatrix}$$

12. Sketch the image and give the new vertices of the rectangle with vertices at $(0, 0)$, $(1, 0)$, $(1, 3)$, and $(0, 3)$ under T , where T is the shear represented by $T(x, y) = (x, y + 2x)$



$$T(0,0) = (0, 0 + 2(0)) = (0,0)$$

$$T(1,0) = (1, 0 + 2(1)) = (1,2)$$

$$T(1,3) = (1, 3 + 2(1)) = (1,5)$$

$$T(0,3) = (0, 3 + 2(0)) = (0,3)$$